Online Appendix: [From Experts to Language Models: Enhancing Online Expert Elicitation with LLMs]

ANONYMOUS AUTHOR(S)

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1 Context Preparations

Table A1. BERT-score semantic fidelity assessment for the initial runs and the final revision.

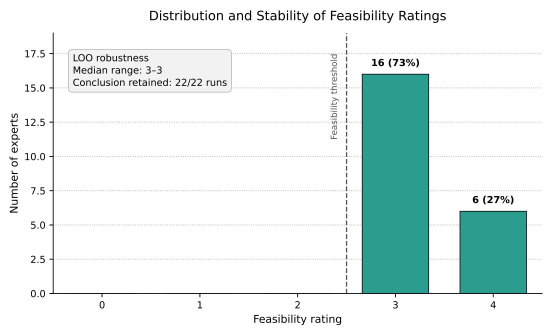
S/No	Raw file	Edited file	Precision	Recall	F1	Status
1	**Section: elicitation**					
	transcription	cleaned_contexts_run01_20260629_104226	0.820	0.823	0.822	not retained
	transcription	cleaned_contexts_run03_20260629_104252	0.827	0.820	0.824	not retained
	transcription	cleaned_contexts_run02_20260629_104239	0.839	0.819	0.829	retained
	author-review-raw	author-review-edited	0.976	0.936	0.956	final version
2.	**Section: project**					
	transcription	cleaned_contexts_run01_20260629_104226	0.862	0.842	0.852	not retained
	transcription	cleaned_contexts_run02_20260629_104239	0.860	0.846	0.853	not retained
	transcription	cleaned_contexts_run03_20260629_104252	0.870	0.845	0.857	retained
	author-review-raw	author-review-edited	0.983	0.968	0.975	final version
3.	**Section: demonstrator**					
	transcription	cleaned_contexts_run02_20260629_104239	0.809	0.842	0.825	not retained
	transcription	cleaned_contexts_run03_20260629_104252	0.834	0.858	0.846	not retained
	transcription	cleaned_contexts_run01_20260629_104226	0.869	0.856	0.862	retained
	author-review-raw	author-review-edited	0.952	0.948	0.950	final version

2 Anchor Phrases

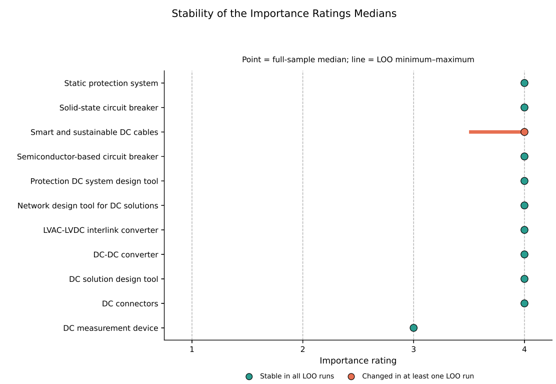
This section presents the literature Sources for Anchor Phrases. It maps literature sources to the phrases used in anchor experiments for both feasibility, importance, and barrier-selection questions. All of the scripts utilized the same anchor phases as stated in the *anchor phrase* column in Table A2.

Table A2. Sources used to support each anchor phrase in the anchor framing prompts.

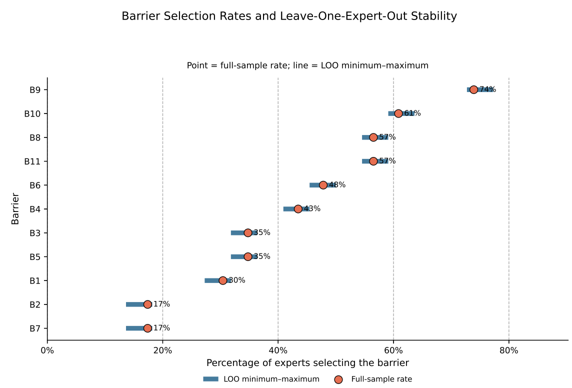
S/No	Anchor type	Anchor phrase	Year	Author/Org.	Citation
1	WORD	European deployments and standardization efforts indicate workable LV/MV-DC microgrids across sectors.	-	-	Derived from other anchor sources
2	EXAMPLE	DC microgrids integrating PV, storage, and DC loads (e.g., ±380 V DC bus) are reported in operational testbeds.	2020	Fraunhofer IISB	[4]
			2014	TU Eindhoven / Vicor	[9]
			2019	EE Power	[5]
			2023	ETSI	[2]
3	NUM_LOW	Some peer-reviewed studies report 2–6% distribution-efficiency improvement with DC microgrids under efficient baselines.	2025	MDPI Energies	[1]
			2015	Aalborg University	[7]
4	NUM_HIGH	Other analyses report 10–18% distribution-efficiency improvement in modeled DC microgrid scenarios with PV and storage.	2023	MDPI Energies	[6]
			2020	Fraunhofer	[3]
			2024	SETIS / EU	[8]



(a) Feasibility at leave-one-out stability check.



(b) Importance ratings at leave-one-out stability check.



(c) Second subfigure description.

Fig. A1. Barrier selection at leave-one-out stability check.

3 Stability Evaluation of DC Experts' Responses—RQ1

This part contains the results of the stability check for experts' responses.

Fig. A2. Predictive posterior checks

4 Posterior predictive check, sensitivity comparisons, and anchor patterns—RQ2

4.1 RQ2 feasibility question

This section presents the results of the Predictive Posterior Check (PPC) and the sensitivity-robustness check for the feasibility question of RQ2. The results are presented through Figure A2 for posterior prediction versus the observed proportions, and the Table A3 for sensitivity comparison between regularizing, primary, and weak priors. Results in Figure A2 show per-model results, which show close alignment between all the predictive posterior and the observed proportions for both the context and the zero-shot. As for the Table A3, the comparisons across the three priors, grouped by model, also show correlations between the Odds Ratios (ORs) and the credible intervals; hence, each model was judged independently as stable, with all inferential conclusions identical across the three priors.

4.2 RQ2 importance question

This section contains the PPC in Table A6 and the sensitivity check in Table A6. But the sensitivity check is only for contrasts of LLM at those Direct Current (DC) solutions whose conclusions were found to be sensitive in the whole run.

Table A4. RQ2 importance: posterior effects of context relative to zero-shot for each base model and DC solution under the primary prior.

Model	DC solution	OR	95% CrI	$P(\mathrm{OR} > 1)$	Interpretation
Gemma3:12B	DC connectors	0.003	[0.001, 0.007]	< 0.001	Lower in Context

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Table A4. RQ2 importance: posterior effects of Context relative to Zero-shot for each base model and DC solution under the primary prior.

Model	DC solution	OR	95% CrI	$P(\text{OR} > 1)$	Interpretation
Gemma3:12B	DC measurement device	0.019	[0.007, 0.043]	< 0.001	Lower in Context
Gemma3:12B	DC solution design tool	3.916	[0.552, 36.900]	0.903	Uncertain
Gemma3:12B	DC-DC converter	0.001	$[4.71 \times 10^{-4}, 0.004]$	< 0.001	Lower in Context
Gemma3:12B	LVAC-LVDC interlink converter	0.014	[0.005, 0.034]	< 0.001	Lower in Context
Gemma3:12B	Network design tool for DC solutions	1.129	[0.688, 1.874]	0.675	Uncertain
Gemma3:12B	Protection DC system design tool	106.152	[33.825, 490.424]	> 0.999	Higher in Context
Gemma3:12B	Semiconductor-based circuit breaker	3.967	[0.539, 37.652]	0.912	Uncertain
Gemma3:12B	Smart and sustainable DC cables	23.185	[5.659, 147.700]	> 0.999	Higher in Context
Gemma3:12B	Solid-state circuit breaker	3.862	[0.530, 41.682]	0.910	Uncertain
Gemma3:12B	Static protection system	1.85×10^3	$[537.585, 8.12 \times 10^3]$	> 0.999	Higher in Context
LLaMa-Pro	DC connectors	9.912	[4.860, 22.219]	> 0.999	Higher in Context
LLaMa-Pro	DC measurement device	0.863	[0.408, 1.747]	0.342	Uncertain
LLaMa-Pro	DC solution design tool	2.544	[1.470, 4.562]	> 0.999	Higher in Context
LLaMa-Pro	DC-DC converter	1.714	[1.110, 2.693]	0.990	Higher in Context
LLaMa-Pro	LVAC-LVDC interlink converter	0.638	[0.360, 1.117]	0.059	Uncertain
LLaMa-Pro	Network design tool for DC solutions	1.709	[1.048, 2.775]	0.987	Higher in Context
LLaMa-Pro	Protection DC system design tool	1.670	[1.048, 2.692]	0.984	Higher in Context
LLaMa-Pro	Semiconductor-based circuit breaker	113.719	[42.297, 398.501]	> 0.999	Higher in Context
LLaMa-Pro	Smart and sustainable DC cables	1.409	[0.893, 2.228]	0.932	Uncertain
LLaMa-Pro	Solid-state circuit breaker	47.547	[22.057, 117.553]	> 0.999	Higher in Context
LLaMa-Pro	Static protection system	41.511	[18.664, 109.765]	> 0.999	Higher in Context
Mistral	DC connectors	0.558	[0.142, 2.152]	0.197	Uncertain
Mistral	DC measurement device	0.126	[0.072, 0.211]	< 0.001	Lower in Context
Mistral	DC solution design tool	5.515	[1.429, 30.078]	0.995	Higher in Context
Mistral	DC-DC converter	0.106	[0.047, 0.218]	< 0.001	Lower in Context

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Table A4. RQ2 importance: posterior effects of Context relative to Zero-shot for each base model and DC solution under the primary prior.

Model	DC solution	OR	95% CrI	$P(\text{OR} > 1)$	Interpretation
Mistral	LVAC-LVDC interlink converter	1.229	[0.773, 1.971]	0.808	Uncertain
Mistral	Network design tool for DC solutions	3.893	[0.533, 40.869]	0.904	Uncertain
Mistral	Protection DC system design tool	9.777	[5.689, 17.474]	> 0.999	Higher in Context
Mistral	Semiconductor-based circuit breaker	205.628	[83.133, 600.933]	> 0.999	Higher in Context
Mistral	Smart and sustainable DC cables	0.126	[0.070, 0.214]	< 0.001	Lower in Context
Mistral	Solid-state circuit breaker	137.006	[44.153, 614.028]	> 0.999	Higher in Context
Mistral	Static protection system	917.910	[297.154, 3.47×10^3]	> 0.999	Higher in Context
Phi-4	DC connectors	0.310	[0.133, 0.662]	0.002	Lower in Context
Phi-4	DC measurement device	0.586	[0.368, 0.924]	0.013	Lower in Context
Phi-4	DC solution design tool	7.443	[1.440, 61.240]	0.993	Higher in Context
Phi-4	DC-DC converter	3.890	[0.544, 38.904]	0.909	Uncertain
Phi-4	LVAC-LVDC interlink converter	7.033	[4.210, 12.203]	> 0.999	Higher in Context
Phi-4	Network design tool for DC solutions	10.745	[2.250, 81.151]	> 0.999	Higher in Context
Phi-4	Protection DC system design tool	10.763	[2.381, 73.933]	> 0.999	Higher in Context
Phi-4	Semiconductor-based circuit breaker	11.671	[2.697, 78.429]	> 0.999	Higher in Context
Phi-4	Smart and sustainable DC cables	3.854	[0.540, 40.953]	0.909	Uncertain
Phi-4	Solid-state circuit breaker	4.701	[0.751, 43.782]	0.948	Uncertain
Phi-4	Static protection system	39.857	[11.877, 219.852]	> 0.999	Higher in Context

Note. Each row represents a separately estimated base-model/DC-solution contrast; the odds ratios were not pooled across DC solutions. OR > 1 indicates higher odds of receiving a higher importance rating in context relative to zero-shot, whereas OR < 1 indicates lower odds in context. The CrI denotes the 95% Bayesian credible interval. $P(\text{OR} > 1)$ denotes the posterior probability that the odds ratio exceeds one.

Table A3. RQ2 feasibility: context vs. zero-shot prior sensitivity comparisons.

Model	Regularizing OR (95% CrI)	Primary OR (95% CrI)	Weak OR (95% CrI)	Conclusions (R/P/W)	Stable
Phi-4	0.220 [0.097, 0.473]	0.139 [0.047, 0.346]	0.117 [0.037, 0.312]	Lower in Context / Lower in Context / Lower in Context	Yes
Gemma3:12B	1.232 [0.391, 3.908]	1.502 [0.292, 8.297]	1.684 [0.290, 11.686]	Uncertain / Uncertain / Uncertain	Yes
Mistral	1.238 [0.384, 3.997]	1.504 [0.306, 8.542]	1.698 [0.275, 12.396]	Uncertain / Uncertain / Uncertain	Yes
LLaMa-Pro	2.979 [1.883, 4.738]	3.254 [2.026, 5.337]	3.329 [2.050, 5.461]	Higher in Context / Higher in Context / Higher in Context	Yes

Note. Cells report posterior median odds ratios with 95% credible intervals. R/P/W denotes regularizing, primary, and weak priors, respectively. Stable indicates that the inferential conclusion was identical under all three priors. OR > 1 indicates higher odds of a higher feasibility rating in Context relative to Zero-shot.

Table A5. RQ2 importance posterior predictive checks

Model	Within 95% PPI	Coverage	Median absolute difference	Maximum absolute difference
Phi-4	66/66	100.0%	0.007	0.013
Gemma3:12B	59/66	89.4%	0.007	0.053
Mistral	66/66	100.0%	0.007	0.033
LLaMa-Pro	66/66	100.0%	0.005	0.020
Overall	257/264	97.3%	0.007	0.053

Note. PPI denotes the 95% posterior predictive interval. Coverage is the proportion of observed rating-category proportions that fell within their corresponding PPIs. Absolute differences compare the observed rating-category proportions with the posterior-predicted median proportions. Each base model contributed 66 checks: 11 DC solutions $\times$ 2 conditions $\times$ 3 observed rating categories. The overall row contains all 264 checks.

Table A6. RQ2 importance contrasts whose conclusions were sensitive to the prior specification.

Model	DC solution	Regularizing OR (95% CrI)	Primary OR (95% CrI)	Weak OR (95% CrI)
Phi-4	DC solution design tool	2.688 [0.916, 8.645]; Uncertain	7.443 [1.440, 61.240]; Higher in Context	19.242 [2.065, 622.900]; Higher in Context
Phi-4	Solid-state circuit breaker	1.980 [0.570, 7.026]; Uncertain	4.701 [0.751, 43.782]; Uncertain	12.196 [1.115, 456.846]; Higher in Context

Note. Cells report the posterior median OR, 95% credible interval, and inferential conclusion under each prior. Only contrasts for which either direction or conclusion changed across priors are shown.

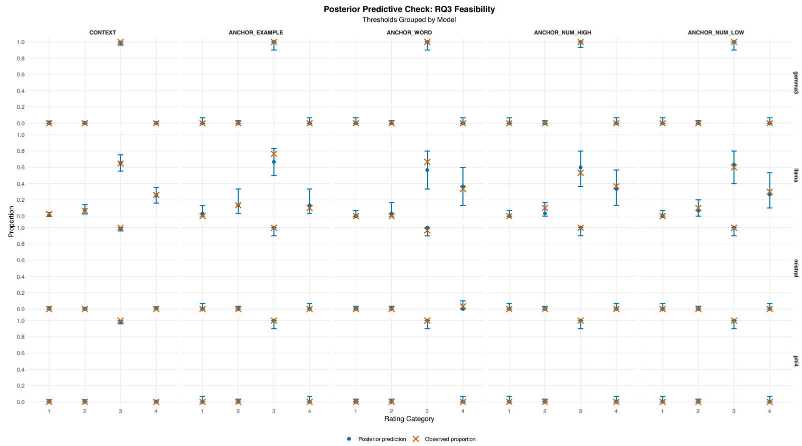

Fig. A3. RQ3 feasibility: posterior predictive checks

5 Posterior predictive check, sensitivity comparisons, and anchor patterns—RQ3

This section

5.0.1 RQ3 feasibility question. This part presents results in two main parts: the PPC and the sensitivity checks.

Table A7. RQ3 feasibility effects of each anchor type relative to context prior-sensitivity comparisons

Model	Anchor type	Regularizing OR (95% CrI)	Primary OR (95% CrI)	Weak OR (95% CrI)	Conclusions (R/P/W)	Stable
Phi-4	Example	1.091 [0.284, 3.934]	1.210 [0.156, 10.258]	1.450 [0.110, 24.053]	Uncertain / Uncertain / Uncertain	Yes
Phi-4	Word	1.067 [0.292, 4.122]	1.218 [0.160, 10.844]	1.455 [0.120, 21.459]	Uncertain / Uncertain / Uncertain	Yes
Phi-4	High numeric	1.067 [0.292, 4.074]	1.208 [0.166, 10.510]	1.457 [0.115, 21.753]	Uncertain / Uncertain / Uncertain	Yes
Phi-4	Low numeric	1.052 [0.280, 4.040]	1.228 [0.151, 10.851]	1.435 [0.123, 22.042]	Uncertain / Uncertain / Uncertain	Yes
Gemma3:12B	Example	1.077 [0.283, 4.131]	1.227 [0.162, 10.743]	1.443 [0.116, 20.737]	Uncertain / Uncertain / Uncertain	Yes
Gemma3:12B	Word	1.076 [0.287, 4.059]	1.226 [0.155, 10.752]	1.501 [0.115, 21.662]	Uncertain / Uncertain / Uncertain	Yes
Gemma3:12B	High numeric	1.069 [0.287, 4.011]	1.257 [0.163, 10.513]	1.462 [0.114, 20.139]	Uncertain / Uncertain / Uncertain	Yes

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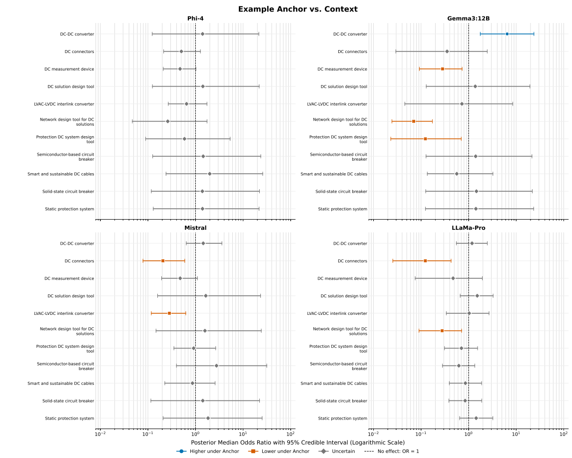
Table A7. Prior-sensitivity comparison for the RQ3 feasibility effects of each anchor type relative to Context.

Model	Anchor type	Regularizing OR (95% CrI)	Primary OR (95% CrI)	Weak OR (95% CrI)	Conclusions (R/P/W)	Stable
Gemma3:12B	Low numeric	1.071 [0.299, 3.887]	1.260 [0.168, 11.360]	1.403 [0.121, 21.725]	Uncertain / Uncertain / Uncertain	Yes
Mistral	Example	1.038 [0.281, 3.766]	1.145 [0.151, 9.070]	1.324 [0.114, 17.059]	Uncertain / Uncertain / Uncertain	Yes
Mistral	Word	1.574 [0.441, 5.722]	3.084 [0.404, 21.888]	5.407 [0.414, 56.770]	Uncertain / Uncertain / Uncertain	Yes
Mistral	High numeric	1.036 [0.291, 3.763]	1.168 [0.160, 9.160]	1.350 [0.120, 18.329]	Uncertain / Uncertain / Uncertain	Yes
Mistral	Low numeric	1.011 [0.268, 3.825]	1.164 [0.156, 8.832]	1.336 [0.116, 19.036]	Uncertain / Uncertain / Uncertain	Yes
LLaMa-Pro	Example	0.569 [0.276, 1.157]	0.525 [0.242, 1.118]	0.506 [0.228, 1.117]	Uncertain / Uncertain / Uncertain	Yes
LLaMa-Pro	Word	1.563 [0.790, 3.056]	1.696 [0.795, 3.658]	1.726 [0.809, 3.678]	Uncertain / Uncertain / Uncertain	Yes
LLaMa-Pro	High numeric	1.420 [0.705, 2.828]	1.509 [0.685, 3.266]	1.524 [0.695, 3.370]	Uncertain / Uncertain / Uncertain	Yes
LLaMa-Pro	Low numeric	1.138 [0.564, 2.231]	1.189 [0.549, 2.523]	1.186 [0.523, 2.647]	Uncertain / Uncertain / Uncertain	Yes

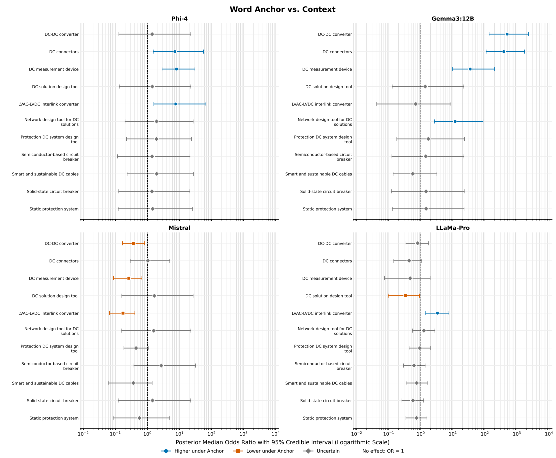
Note. Cells report posterior median odds ratios with 95% credible intervals. R/P/W denotes regularizing, primary, and weak priors, respectively. Stable indicates that the inferential conclusion was identical under all three priors. OR $>$ 1 indicates higher odds of a higher feasibility rating under the specified anchor relative to Context.

5.1 RQ3 importance question

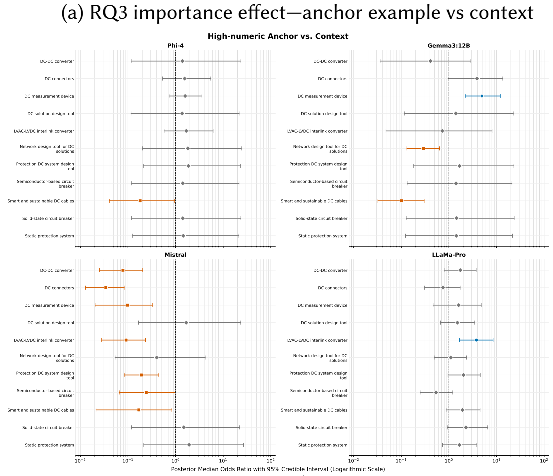
This part presents



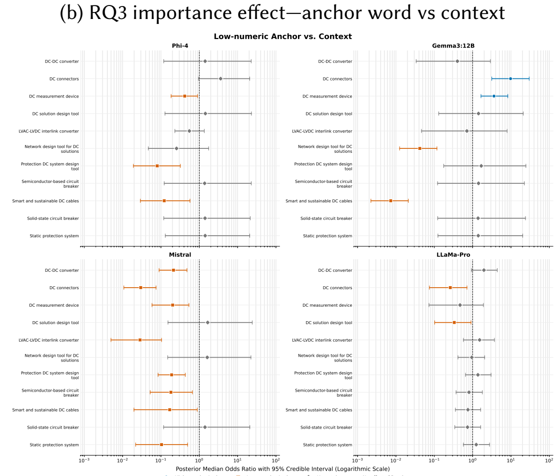
(a) RQ3 importance effect—anchor example vs context



(b) RQ3 importance effect—anchor word vs context



(c) RQ3 importance effect—anchor high number vs context



(d) RQ3 importance effect—anchor low number vs context

Fig. A4. RQ3 importance effect on primary priors for each anchor vs context

Table A8. RQ3 importance: primary model posterior predictive checks

Model	Comparison group	Within 95% PPI	Coverage	Med.abs difference	Max.abs difference
Phi-4	Context	22/22	100.0%	0.007	0.013
Phi-4	Example anchor	22/22	100.0%	0.000	0.033
Phi-4	Word anchor	22/22	100.0%	0.000	0.033
Phi-4	High-numeric anchor	22/22	100.0%	0.000	0.033
Phi-4	Low-numeric anchor	22/22	100.0%	0.000	0.067
Gemma3:12B	Context	22/22	100.0%	0.007	0.027
Gemma3:12B	Example anchor	22/22	100.0%	0.000	0.033
Gemma3:12B	Word anchor	22/22	100.0%	0.000	0.067
Gemma3:12B	High-numeric anchor	22/22	100.0%	0.000	0.033
Gemma3:12B	Low-numeric anchor	22/22	100.0%	0.000	0.067
Mistral	Context	22/22	100.0%	0.013	0.027
Mistral	Example anchor	22/22	100.0%	0.000	0.033
Mistral	Word anchor	22/22	100.0%	0.000	0.033
Mistral	High-numeric anchor	22/22	100.0%	0.033	0.033
Mistral	Low-numeric anchor	22/22	100.0%	0.033	0.033
LLaMa-Pro	Context	22/22	100.0%	0.003	0.013
LLaMa-Pro	Example anchor	22/22	100.0%	0.000	0.033
LLaMa-Pro	Word anchor	22/22	100.0%	0.000	0.017
LLaMa-Pro	High-numeric anchor	22/22	100.0%	0.000	0.033
LLaMa-Pro	Low-numeric anchor	22/22	100.0%	0.000	0.000
Overall	Overall	440/440	100.0%	0.000	0.067

Note. PPI denotes the 95% posterior predictive interval. Coverage is the proportion of observed rating-category proportions within their corresponding PPIs. Absolute differences compare observed proportions with posterior-predicted median proportions.

Table A9. Prior-sensitivity summary for the RQ3 importance anchor effects.

Model	Anchor type	Contrasts	Direction stable	Conclusion stable
Phi-4	Example	11	11/11 (100.0%)	11/11 (100.0%)
Phi-4	Word	11	11/11 (100.0%)	10/11 (90.9%)
Phi-4	High numeric	11	11/11 (100.0%)	10/11 (90.9%)
Phi-4	Low numeric	11	11/11 (100.0%)	10/11 (90.9%)
Gemma3:12B	Example	11	11/11 (100.0%)	9/11 (81.8%)
Gemma3:12B	Word	11	11/11 (100.0%)	11/11 (100.0%)
Gemma3:12B	High numeric	11	11/11 (100.0%)	10/11 (90.9%)
Gemma3:12B	Low numeric	11	11/11 (100.0%)	11/11 (100.0%)
Mistral	Example	11	10/11 (90.9%)	10/11 (90.9%)
Mistral	Word	11	10/11 (90.9%)	10/11 (90.9%)
Mistral	High numeric	11	11/11 (100.0%)	9/11 (81.8%)
Mistral	Low numeric	11	11/11 (100.0%)	10/11 (90.9%)
LLaMa-Pro	Example	11	10/11 (90.9%)	11/11 (100.0%)
LLaMa-Pro	Word	11	11/11 (100.0%)	9/11 (81.8%)
LLaMa-Pro	High numeric	11	11/11 (100.0%)	11/11 (100.0%)
LLaMa-Pro	Low numeric	11	11/11 (100.0%)	10/11 (90.9%)
Overall	All	176	173/176 (98.3%)	162/176 (92.0%)

Note. Direction stability indicates that the posterior median OR remained on the same side of one under the regularizing, primary, and weak priors. Conclusion stability indicates that the inferential classification was identical under all three priors. Rates were calculated directly from the 176 contrast-level records.

Table A10. RQ3 importance contrasts with prior-dependent inferential conclusions.

Model	Anchor type	DC solution	Regularizing	Primary	Weak
Phi-4	Word	DC connectors	2.764 [0.982, 8.945]; Uncertain	7.135 [1.515, 55.912]; Higher under Anchor	18.259 [2.161, 612.043]; Higher under Anchor
Phi-4	High numeric	Smart and sustainable DC cables	0.437 [0.140, 1.417]; Uncertain	0.180 [0.041, 0.957]; Lower under Anchor	0.104 [0.017, 0.657]; Lower under Anchor
Phi-4	Low numeric	DC connectors	2.092 [0.805, 6.264]; Uncertain	3.611 [0.956, 20.708]; Uncertain	5.046 [1.110, 51.317]; Higher under Anchor
Gemma3:12B	Example	DC-DC converter	2.366 [0.892, 6.010]; Uncertain	6.462 [1.759, 23.601]; Higher under Anchor	12.988 [2.815, 62.114]; Higher under Anchor
Gemma3:12B	Example	Protection DC system design tool	0.343 [0.102, 1.237]; Uncertain	0.124 [0.023, 0.700]; Lower under Anchor	0.073 [0.009, 0.496]; Lower under Anchor
Gemma3:12B	High numeric	DC connectors	1.636 [0.595, 4.111]; Uncertain	3.920 [0.959, 13.521]; Uncertain	7.591 [1.633, 35.109]; Higher under Anchor
Mistral	Example	DC connectors	0.426 [0.191, 1.019]; Uncertain	0.207 [0.079, 0.594]; Lower under Anchor	0.146 [0.050, 0.445]; Lower under Anchor
Mistral	Example	Protection DC system design tool	1.045 [0.466, 2.527]; Uncertain	0.913 [0.352, 2.654]; Uncertain	0.852 [0.325, 2.733]; Uncertain
Mistral	Word	DC connectors	1.325 [0.497, 4.001]; Uncertain	1.030 [0.290, 4.940]; Uncertain	0.815 [0.195, 5.322]; Uncertain
Mistral	Word	Protection DC system design tool	0.579 [0.272, 1.311]; Uncertain	0.440 [0.184, 1.097]; Uncertain	0.392 [0.164, 0.995]; Lower under Anchor
Mistral	High numeric	Semiconductor-based circuit breaker	0.436 [0.154, 1.342]; Uncertain	0.242 [0.066, 0.974]; Lower under Anchor	0.189 [0.045, 0.861]; Lower under Anchor
Mistral	High numeric	Smart and sustainable DC cables	0.437 [0.136, 1.281]; Uncertain	0.169 [0.021, 0.836]; Lower under Anchor	0.067 [0.002, 0.632]; Lower under Anchor
Mistral	Low numeric	Smart and sustainable DC cables	0.437 [0.131, 1.254]; Uncertain	0.169 [0.020, 0.884]; Lower under Anchor	0.067 [0.002, 0.601]; Lower under Anchor

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Table A10. RQ3 importance contrasts with prior-dependent inferential conclusions.

Model	Anchor type	DC solution	Regularizing	Primary	Weak
LLaMa-Pro	Example	LVAC-LVDC interlink converter	0.932 [0.375, 2.159]; Uncertain	1.028 [0.340, 2.686]; Uncertain	1.070 [0.337, 2.860]; Uncertain
LLaMa-Pro	Word	DC connectors	0.560 [0.241, 1.223]; Uncertain	0.431 [0.143, 1.063]; Uncertain	0.391 [0.131, 0.974]; Lower under Anchor
LLaMa-Pro	Word	DC solution design tool	0.458 [0.179, 1.058]; Uncertain	0.329 [0.096, 0.911]; Lower under Anchor	0.289 [0.073, 0.849]; Lower under Anchor
LLaMa-Pro	Low numeric	DC solution design tool	0.461 [0.184, 1.033]; Uncertain	0.332 [0.103, 0.907]; Lower under Anchor	0.286 [0.071, 0.850]; Lower under Anchor

Note. Cells report posterior median ORs, 95% credible intervals, and inferential conclusions. Only contrasts whose direction or conclusion differed across the regularizing, primary, and weak priors are listed.

Table A11. RQ3 importance: primary-prior upper-end probability contrasts for the word and example anchors.

Model	DC solution	Anchor	Context $P(Y = 4)$	Anchor $P(Y = 4)$	Δ	95% CrI	$P(\Delta > 0)$	Interpretation
Gemma3:12B	DC connectors	Word	0.029	0.919	+0.888	[0.767, 0.949]	> 0.999	Greater
Gemma3:12B	DC connectors	Example	0.029	0.010	-0.017	[-0.048, 0.035]	0.160	Uncertain
Gemma3:12B	DC measurement device	Word	0.376	0.954	+0.574	[0.456, 0.659]	> 0.999	Greater
Gemma3:12B	DC measurement device	Example	0.376	0.147	-0.228	[-0.348, -0.068]	0.004	Reduced
Gemma3:12B	DC solution design tool	Word	0.996	0.997	+0.001	[-0.022, 0.010]	0.596	Uncertain
Gemma3:12B	DC solution design tool	Example	0.996	0.997	+0.001	[-0.023, 0.010]	0.597	Uncertain
Gemma3:12B	DC-DC converter	Word	0.023	0.916	+0.891	[0.770, 0.955]	> 0.999	Greater
Gemma3:12B	DC-DC converter	Example	0.023	0.131	+0.107	[0.021, 0.256]	0.997	Greater
Gemma3:12B	LVAC-LVDC interlink converter	Word	0.004	0.002	-0.001	[-0.010, 0.026]	0.398	Uncertain
Gemma3:12B	LVAC-LVDC interlink converter	Example	0.004	0.003	-0.001	[-0.010, 0.024]	0.406	Uncertain
Gemma3:12B	Network design tool for DC solutions	Word	0.738	0.971	+0.227	[0.133, 0.305]	> 0.999	Greater
Gemma3:12B	Network design tool for DC solutions	Example	0.738	0.165	-0.570	[-0.692, -0.409]	< 0.001	Reduced
Gemma3:12B	Protection DC system design tool	Word	0.992	0.995	+0.002	[-0.029, 0.017]	0.667	Uncertain
Gemma3:12B	Protection DC system design tool	Example	0.992	0.940	-0.051	[-0.158, -0.005]	0.009	Reduced
Gemma3:12B	Semiconductor-based circuit breaker	Word	0.996	0.997	+0.001	[-0.025, 0.010]	0.602	Uncertain
Gemma3:12B	Semiconductor-based circuit breaker	Example	0.996	0.997	+0.001	[-0.023, 0.010]	0.601	Uncertain
Gemma3:12B	Smart and sustainable DC cables	Word	0.971	0.950	-0.020	[-0.124, 0.026]	0.240	Uncertain
Gemma3:12B	Smart and sustainable DC cables	Example	0.971	0.950	-0.020	[-0.124, 0.027]	0.247	Uncertain
Gemma3:12B	Solid-state circuit breaker	Word	0.996	0.998	+0.001	[-0.024, 0.011]	0.606	Uncertain
Gemma3:12B	Solid-state circuit breaker	Example	0.996	0.998	+0.001	[-0.023, 0.011]	0.608	Uncertain
Gemma3:12B	Static protection system	Word	0.996	0.997	+0.001	[-0.026, 0.010]	0.615	Uncertain
Gemma3:12B	Static protection system	Example	0.996	0.997	+0.001	[-0.025, 0.011]	0.605	Uncertain
LLaMa-Pro	DC connectors	Word	0.326	0.172	-0.153	[-0.284, 0.013]	0.033	Uncertain
LLaMa-Pro	DC connectors	Example	0.326	0.057	-0.266	[-0.356, -0.146]	< 0.001	Reduced
LLaMa-Pro	DC measurement device	Word	0.077	0.038	-0.038	[-0.094, 0.056]	0.165	Uncertain
LLaMa-Pro	DC measurement device	Example	0.077	0.038	-0.038	[-0.093, 0.056]	0.159	Uncertain

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Table A11. RQ3 importance: primary-prior upper-end probability contrasts for the word and example anchors.

Model	DC solution	Anchor	Context $P(Y=4)$	Anchor $P(Y=4)$	Δ	95% CrI	$P(\Delta > 0)$	Interpretation
LLaMa-Pro	DC solution design tool	Word	0.269	0.109	-0.159	[-0.262, -0.016]	0.014	Reduced
LLaMa-Pro	DC solution design tool	Example	0.269	0.357	+0.089	[-0.077, 0.268]	0.848	Uncertain
LLaMa-Pro	DC-DC converter	Word	0.424	0.367	-0.055	[-0.231, 0.133]	0.271	Uncertain
LLaMa-Pro	DC-DC converter	Example	0.424	0.464	+0.041	[-0.138, 0.221]	0.669	Uncertain
LLaMa-Pro	LVAC-LVDC interlink converter	Word	0.157	0.378	+0.221	[0.056, 0.406]	0.997	Greater
LLaMa-Pro	LVAC-LVDC interlink converter	Example	0.157	0.160	+0.004	[-0.111, 0.160]	0.519	Uncertain
LLaMa-Pro	Network design tool for DC solutions	Word	0.377	0.429	+0.051	[-0.133, 0.244]	0.707	Uncertain
LLaMa-Pro	Network design tool for DC solutions	Example	0.377	0.145	-0.231	[-0.350, -0.072]	0.004	Reduced
LLaMa-Pro	Protection DC system design tool	Word	0.415	0.398	-0.016	[-0.189, 0.168]	0.432	Uncertain
LLaMa-Pro	Protection DC system design tool	Example	0.415	0.334	-0.080	[-0.243, 0.106]	0.194	Uncertain
LLaMa-Pro	Semiconductor-based circuit breaker	Word	0.688	0.575	-0.112	[-0.294, 0.065]	0.112	Uncertain
LLaMa-Pro	Semiconductor-based circuit breaker	Example	0.688	0.577	-0.109	[-0.299, 0.063]	0.112	Uncertain
LLaMa-Pro	Smart and sustainable DC cables	Word	0.541	0.469	-0.071	[-0.254, 0.122]	0.240	Uncertain
LLaMa-Pro	Smart and sustainable DC cables	Example	0.541	0.501	-0.039	[-0.226, 0.151]	0.338	Uncertain
LLaMa-Pro	Solid-state circuit breaker	Word	0.678	0.543	-0.134	[-0.322, 0.041]	0.072	Uncertain
LLaMa-Pro	Solid-state circuit breaker	Example	0.678	0.639	-0.038	[-0.224, 0.130]	0.331	Uncertain
LLaMa-Pro	Static protection system	Word	0.581	0.508	-0.073	[-0.259, 0.105]	0.213	Uncertain
LLaMa-Pro	Static protection system	Example	0.581	0.665	+0.084	[-0.106, 0.248]	0.812	Uncertain
Mistral	DC connectors	Word	0.939	0.941	+0.002	[-0.106, 0.062]	0.515	Uncertain
Mistral	DC connectors	Example	0.939	0.763	-0.175	[-0.342, -0.045]	0.003	Reduced
Mistral	DC measurement device	Word	0.396	0.147	-0.247	[-0.368, -0.088]	0.002	Reduced
Mistral	DC measurement device	Example	0.396	0.238	-0.157	[-0.301, 0.022]	0.040	Uncertain
Mistral	DC solution design tool	Word	0.993	0.996	+0.002	[-0.030, 0.016]	0.649	Uncertain
Mistral	DC solution design tool	Example	0.993	0.996	+0.002	[-0.029, 0.016]	0.652	Uncertain
Mistral	DC-DC converter	Word	0.653	0.412	-0.241	[-0.418, -0.049]	0.006	Reduced
Mistral	DC-DC converter	Example	0.653	0.732	+0.079	[-0.102, 0.231]	0.807	Uncertain

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Table A11. RQ3 importance: primary-prior upper-end probability contrasts for the word and example anchors.

Model	DC solution	Anchor	Context $P(Y = 4)$	Anchor $P(Y = 4)$	Δ	95% CrI	$P(\Delta > 0)$	Interpretation
Mistral	LVAC-LVDC interlink converter	Word	0.619	0.219	-0.398	[-0.543, -0.222]	< 0.001	Reduced
Mistral	LVAC-LVDC interlink converter	Example	0.619	0.316	-0.303	[-0.466, -0.116]	0.001	Reduced
Mistral	Network design tool for DC solutions	Word	0.994	0.996	+0.001	[-0.028, 0.014]	0.635	Uncertain
Mistral	Network design tool for DC solutions	Example	0.994	0.996	+0.001	[-0.026, 0.014]	0.638	Uncertain
Mistral	Protection DC system design tool	Word	0.853	0.717	-0.134	[-0.311, 0.013]	0.038	Uncertain
Mistral	Protection DC system design tool	Example	0.853	0.841	-0.012	[-0.165, 0.100]	0.431	Uncertain
Mistral	Semiconductor-based circuit breaker	Word	0.973	0.990	+0.015	[-0.035, 0.044]	0.819	Uncertain
Mistral	Semiconductor-based circuit breaker	Example	0.973	0.990	+0.015	[-0.033, 0.045]	0.826	Uncertain
Mistral	Smart and sustainable DC cables	Word	0.108	0.042	-0.064	[-0.127, 0.031]	0.073	Uncertain
Mistral	Smart and sustainable DC cables	Example	0.108	0.094	-0.013	[-0.099, 0.117]	0.406	Uncertain
Mistral	Solid-state circuit breaker	Word	0.996	0.998	+0.001	[-0.024, 0.011]	0.606	Uncertain
Mistral	Solid-state circuit breaker	Example	0.996	0.997	+0.001	[-0.026, 0.011]	0.604	Uncertain
Mistral	Static protection system	Word	0.990	0.981	-0.007	[-0.078, 0.014]	0.298	Uncertain
Mistral	Static protection system	Example	0.990	0.994	+0.004	[-0.034, 0.022]	0.693	Uncertain
Phi-4	DC connectors	Word	0.855	0.977	+0.118	[0.038, 0.184]	0.995	Greater
Phi-4	DC connectors	Example	0.855	0.748	-0.106	[-0.275, 0.032]	0.071	Uncertain
Phi-4	DC measurement device	Word	0.556	0.910	+0.350	[0.213, 0.456]	> 0.999	Greater
Phi-4	DC measurement device	Example	0.556	0.373	-0.182	[-0.355, 0.005]	0.029	Uncertain
Phi-4	DC solution design tool	Word	0.996	0.997	+0.001	[-0.023, 0.011]	0.608	Uncertain
Phi-4	DC solution design tool	Example	0.996	0.997	+0.001	[-0.024, 0.011]	0.603	Uncertain
Phi-4	DC-DC converter	Word	0.996	0.997	+0.001	[-0.025, 0.010]	0.601	Uncertain
Phi-4	DC-DC converter	Example	0.996	0.998	+0.001	[-0.025, 0.011]	0.606	Uncertain
Phi-4	LVAC-LVDC interlink converter	Word	0.843	0.976	+0.129	[0.047, 0.196]	0.995	Greater
Phi-4	LVAC-LVDC interlink converter	Example	0.843	0.777	-0.065	[-0.231, 0.070]	0.188	Uncertain
Phi-4	Network design tool for DC solutions	Word	0.990	0.995	+0.004	[-0.033, 0.021]	0.694	Uncertain
Phi-4	Network design tool for DC solutions	Example	0.990	0.961	-0.027	[-0.121, 0.006]	0.078	Uncertain

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Table A11. RQ3 importance: primary-prior upper-end probability contrasts for the word and example anchors.

Model	DC solution	Anchor	Context $P(Y = 4)$	Anchor $P(Y = 4)$	Δ	95% CrI	$P(\Delta > 0)$	Interpretation
Phi-4	Protection DC system design tool	Word	0.989	0.994	+0.004	[-0.033, 0.022]	0.707	Uncertain
Phi-4	Protection DC system design tool	Example	0.989	0.981	-0.007	[-0.078, 0.015]	0.304	Uncertain
Phi-4	Semiconductor-based circuit breaker	Word	0.996	0.997	+0.001	[-0.022, 0.010]	0.602	Uncertain
Phi-4	Semiconductor-based circuit breaker	Example	0.996	0.998	+0.001	[-0.024, 0.010]	0.597	Uncertain
Phi-4	Smart and sustainable DC cables	Word	0.987	0.994	+0.005	[-0.033, 0.025]	0.717	Uncertain
Phi-4	Smart and sustainable DC cables	Example	0.987	0.994	+0.005	[-0.033, 0.025]	0.717	Uncertain
Phi-4	Solid-state circuit breaker	Word	0.996	0.997	+0.001	[-0.025, 0.010]	0.606	Uncertain
Phi-4	Solid-state circuit breaker	Example	0.996	0.997	+0.001	[-0.023, 0.010]	0.611	Uncertain
Phi-4	Static protection system	Word	0.996	0.998	+0.001	[-0.025, 0.011]	0.605	Uncertain
Phi-4	Static protection system	Example	0.996	0.997	+0.001	[-0.023, 0.010]	0.603	Uncertain

Note. Δ is the posterior anchor-minus-Context difference in $P(Y = 4)$. Only ratings 3 and 4 were observed in the importance data; consequently, $P(Y = 1)$ was not estimated by the fitted importance model.

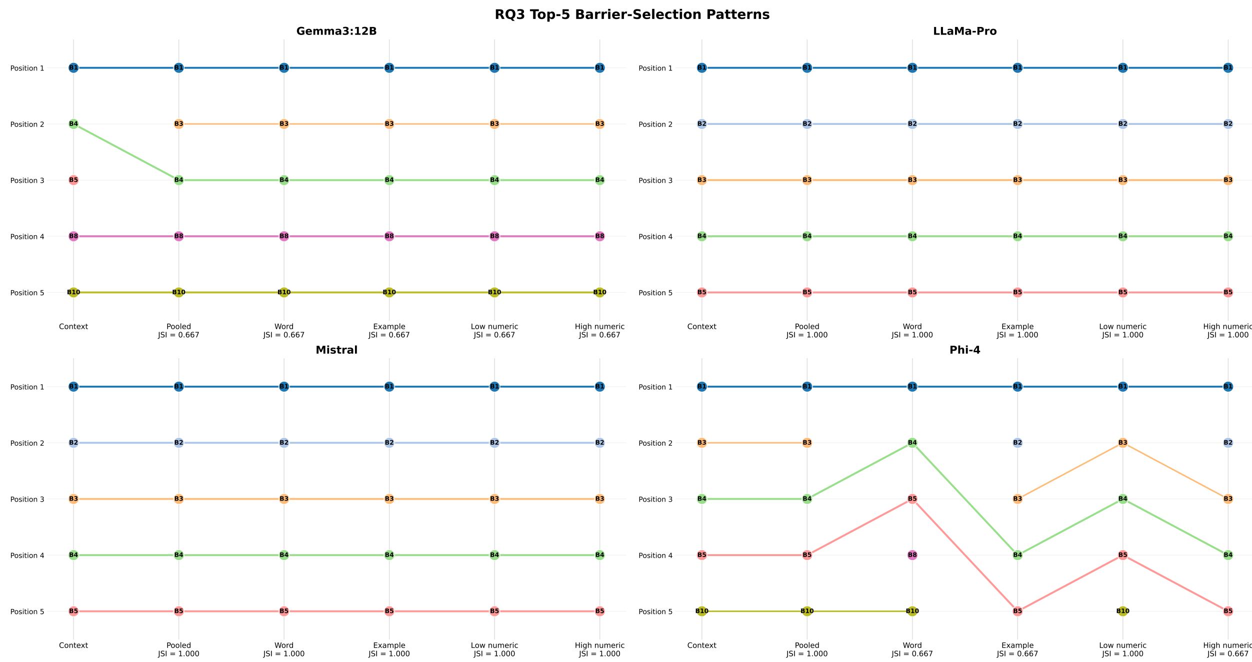

Fig. A5. RQ3 barrier selection pattern between—each anchor arm vs context

Fig. A6. Lines connect the same barrier while it remains in the top-five set. A line that begins or ends indicates entry into or removal from the set. Display positions are ordered by barrier ID, not by selection rank.

5.2 Barrier selection—RQ3

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